Algebra Mastery Pack — Practice Test C

Grade 11 algebra, functions & data analysis · 24 questions · suggested time 45 minutes · no calculator needed

Name: _____ Date: _____
Score: _____ / 24

Instructions. Write full working in the space under each question; answers alone do not show how you got there. Circle one letter for a multiple-choice question and write a single value on the answer line for a short-answer question. Answer every question — a blank scores zero. The answer key with a worked solution for every question is a separate PDF; mark your own paper only after you finish.

1. [Solving Linear Equations (for x)]

Solve $7x = 42$. Enter the value of x.

Answer: _____

2. [Literal Equations & Solving for y]

Solve $P = 2l + 2w$ for w.

A) $w = (P - 2l) \div 2$
B) $w = P - 2l$
C) $w = (P + 2l) \div 2$
D) $w = 2P - 2l$

3. [Linear Inequalities]

Solve $3x - 6 \geq 9$.

A) $x \geq 5$
B) $x \leq 5$
C) $x \geq 15$
D) $x \leq 15$

4. [Systems of Linear Equations]

Solve by elimination: $2x + y = 8$ and $x - y = 1$.

A) (3, 2)
B) (2, 3)
C) (3, 3)
D) (1, 6)

5. [Functions: Notation, Domain & Range]

If $h(x) = 5x - 4$, what is $h(2)$? Enter a number.

Answer: _____

6. [Linear Functions: Slope, Intercepts & Equations]

Find the slope through the points (1, 2) and (3, 8). Enter a number.

Answer: _____

7. [Quadratic Functions & Equations]

The graph of $y = -x^2 + 3$ opens ___.

A) downward
B) upward
C) left
D) right

8. [Polynomials: Operations & Factoring]

Factor out the greatest common factor from $6x + 12$. Enter the GCF.

Answer: _____

9. [Radical & Rational Expressions]

To solve a rational equation, multiply both sides by the ___.

A) least common denominator (LCD)
B) greatest common factor
C) numerator
D) variable

10. [Exponential Functions & Growth/Decay]

For $y = 100 \cdot (1/2)^x$, find y when x = 2. Enter a number.

Answer: _____

11. [Data: Measures of Center & Spread]

Which measure of center is most affected by an outlier (one very large or small value)?

A) Mean
B) Median
C) Mode
D) Range

12. [Scatter Plots, Line of Best Fit & Correlation]

A correlation coefficient r near -1 indicates ___.

A) strong negative correlation
B) strong positive correlation
C) no correlation
D) an error

13. [Absolute Value Equations & Inequalities]

Solve $|2x + 1| = 9$. Enter the larger of the two solutions.

Answer: _____

14. [Arithmetic & Geometric Sequences]

What is the common ratio of the geometric sequence 2, 6, 18, 54, ...?

A) 2
B) 3
C) 4
D) 12

15. [Inverse Functions]

For $f(x) = x^2$ with domain $x \geq 0$, what is $f^{-1}(x)$?

A) $\sqrt{x}$
B) x^2
C) $1/x^2$
D) $-\sqrt{x}$

16. [Function Transformations & Vertex Form]

What does $y = -f(x)$ do to the graph of $y = f(x)$?

A) reflects it across the x-axis
B) reflects it across the y-axis
C) shifts it 1 unit down
D) rotates it 90°

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17. [Systems of Linear Inequalities & Linear Programming]

A feasible region has vertices (0, 0), (5, 0), (0, 4) and (2, 3). Where is $P = 3x + 2y$ maximised?

A) (0, 0)
B) (5, 0)
C) (0, 4)
D) (2, 3)

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18. [Probability from Two-Way Tables]

Survey of 200 students: how they travel and whether they arrived on time

	On time	Late	Total
Rides the bus	60	20	80
Walks	90	30	120
Total	150	50	200

What is P(walks and is late)?

A) 0.15
B) 0.25
C) 0.60
D) 0.30

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19. [Quadratic Functions & Equations]

For $y = x^2 - 4x + 3$, the vertex's x-coordinate is $x = -b \div (2a)$. Enter that value.

Answer: _____

20. [Absolute Value Equations & Inequalities]

Solve $|x - 4| = -2$.

A) $x = 6$
B) $x = 2$
C) $x = 6$ or $x = 2$
D) No solution

21. [Arithmetic & Geometric Sequences]

A geometric sequence starts 5, 10, 20, … What is its 6th term?

Answer: _____

22. [Inverse Functions]

Which function has an inverse that is also a function?

A) $f(x) = x^2$ for all real x
B) $f(x) = |x|$ for all real x
C) $f(x) = 2x + 5$
D) $f(x) = 4$ (a constant)

23. [Function Transformations & Vertex Form]

$y = f(2x)$ is a…

A) horizontal compression by a factor of 1/2
B) horizontal stretch by a factor of 2
C) vertical compression by 1/2
D) vertical stretch by 2

24. [Systems of Linear Inequalities & Linear Programming]

A region is bounded by $x \geq 0$, $y \geq 0$ and $x + y \leq 10$. What is the largest value of $P = 2x + 3y$ on that region?

Answer: _____